Boing Blockchain — Implementation Roadmap

> ***Priorities:*** *Security → Scalability → Decentralization → Authenticity*
> ***Goal:*** *Build an authentic, efficient L1 with unique innovations.*
> ***Aligned with:*** *BOING-BLOCKCHAIN-DESIGN-PLAN.md (UX, Technical Innovations, Sustainability)*
> ***Recommendations:*** *DEVELOPMENT-AND-ENHANCEMENTS.md (SDK, Automation, dApp Incentives)*

Quick Start

```
# Build the workspace
cargo build

# Run the node
cargo run -p boing-node

# Run tests
cargo test
```

Project Structure

```
boing-network/
├── Cargo.toml              # Workspace root
├── crates/
│   ├── boing-primitives/   # Types, hash, crypto, signatures
│   ├── boing-consensus/    # PoS + HotStuff BFT
│   ├── boing-state/        # Verkle tree, state store
│   ├── boing-execution/    # VM, parallel scheduler
│   ├── boing-governance/   # Phased governance (time locks)
│   ├── boing-automation/   # Scheduler, triggers, executor incentives
│   ├── boing-cli/          # boing init, dev, deploy
│   ├── boing-p2p/          # libp2p networking
│   └── boing-node/         # Node binary
├── docs/                   # All project documentation
├── fuzz/                   # Cargo-fuzz harness (boing-primitives)
└── website/                # boing.network site (Astro + Cloudflare)
```

Phase 1: Foundation (Weeks 1–8)

1.1 Primitives — COMPLETE ✓

- ☑ `Hash`, `Hasher` (BLAKE3)
- ☑ `AccountId`, `Account`, `AccountState`
- ☑ `Transaction`, `TransactionPayload`, `AccessList`
- ☑ `Block`, `BlockHeader`
- ☑ Ed25519 signature; `SignedTransaction`; sign/verify
- ☑ Unit tests (hash, tx id, signatures, access-list conflicts)

1.2 Consensus — COMPLETE ✓

- [x] HotStuff state machine (Propose → Vote → Commit)
- [x] Quorum (2f+1); single-validator and multi-validator modes
- [x] Unit tests (propose_and_commit, propose + vote flow)
- [x] Leader rotation (round-robin)
- [x] Simulate 4 nodes, 1 Byzantine

1.3 State Store

- [x] In-memory `StateStore` (HashMap)
- [x] Sparse Merkle tree for state root
- [x] Transaction root (Merkle tree of tx IDs via `tx_root`)
- [x] State root from Sparse Merkle
- [x] Apply/revert batch (checkpoint/revert)

1.4 Execution (Minimal)

- [x] VM: `Transfer` execution
- [x] `TransactionScheduler` (access-list batching)
- [x] Unit tests (scheduler batches)
- [x] `BlockExecutor` (execute block of txs)
- [x] Gas metering stub (fixed per tx type)
- [x] Nonce validation

Phase 2: Networking & Integration (Weeks 9–16)

2.1 P2P Layer

- [x] libp2p swarm setup (TCP, noise, yamux, TLS)
- [x] Gossip: blocks, transactions (gossipsub)
- [x] Peer discovery (mDNS)
- [x] Request/response: get blocks by hash/height
- [x] Block propagation and import (full node receives via gossip)

2.2 Advanced Decentralized Peer Discovery (see DECENTRALIZATION-AND-NETWORKING.md)

- [ ] DHT-based discovery (Kademlia); minimize fixed bootnode reliance
- [ ] Gossip-first overlay: random peer selection, active probing
- [ ] Bootnode rotation (governance or community-funded)
- [ ] Peer scoring/reputation for Sybil/eclipse resistance
- [ ] WebRTC/WebSockets for browser light clients; decentralized signaling
- [ ] Incentivized relayers; DHT rendezvous for NAT traversal

2.2b Decentralized WebRTC Signaling (see DECENTRALIZATION-AND-NETWORKING.md)

- [ ] Signaling smart contract (offer/answer exchange, event emission)
- [ ] Spam prevention: on-chain rate limiting, deposit/stake for offers, identity/reputation gate

- [ ] IPFS/Filecoin integration for large SDP payloads; on-chain CID pointers
- [ ] DHT announcement of WebRTC capability; recipient lookup
- [ ] Incentivized STUN/TURN servers; protocol rewards for relay providers
- [ ] STUN/TURN on-chain registry with staking; reputation score from uptime, latency, success rate
- [ ] STUN/TURN slashing for poor performance; SDK server selection by reputation

2.3 Node Integration

- [x] Mempool (transaction pool)
- [x] Wire consensus ↔ execution ↔ state
- [x] Block production loop (BlockProducer + produce_block_if_ready)
- [x] Chain state (ChainState, genesis, append)
- [x] CLI: `--validator`, `--rpc-port`, `--data-dir`
- [x] JSON-RPC HTTP server (boing_submitTransaction, boing_chainHeight)
- [x] Validator loop (produce blocks every BLOCK_TIME when --validator)
- [x] Block import and validation (validate_and_execute_block, import_block)

2.4 End-to-End

- [x] Single node: produce blocks with txs
- [x] Two nodes: sync blocks (P2P gossip + block import)
- [x] Multi-node testnet (4+ validators)

Phase 3: Verkle & VM (Weeks 17–28)

3.1 Verkle Tree / Sparse Merkle

- [x] Sparse Merkle tree (insert, get, delete, root)
- [x] State root from Sparse Merkle
- [x] Proof generation (MerkleProof, prove/verify)
- [x] Stateless client verification (boing_getAccountProof, boing_verifyAccountProof RPC)

3.2 Custom VM

- [x] Bytecode spec (PUSH, ADD, SUB, MUL, MLOAD, MSTORE, SLOAD, SSTORE, JUMP, JUMPI, RETURN, STOP)
- [x] Interpreter (deterministic stack machine)
- [x] Gas metering per opcode
- [x] Contract deployment and call

3.3 Parallel Execution (Full)

- [x] Execute Transfer-only batches in parallel (rayon)
- [x] Conflict detection at runtime (sanity check)
- [x] Benchmarks: sequential vs parallel (criterion)

Phase 4: PoS & Production Readiness (Weeks 29–40)

4.1 Proof of Stake

- [x] Validator set from stake (`StateStore::top_stakers`)
- [x] Staking transactions (bond, unbond)
- [x] Slashing conditions (equivocation detection in consensus)
- [x] Rewards distribution (block emission to proposer)
- [x] Sustainable tokenomics (design + boing-tokenomics crate: emission curve, fee split, burn, dApp cap)

4.2 Core Innovations (from Design Plan)

- [ ] Native Account Abstraction (protocol-level)
- [x] Adaptive gas model (GasConfig, multiplier with predictable caps)
- [x] Phased governance (boing-governance: proposal → cooling → execution)
- [x] Transparent slashing with appeal

4.3 Hardening

- [x] Fuzz testing (cargo-fuzz; fuzz/ crate, bincode Block/Transaction)
- [x] Property-based tests (proptest: balance preservation, parallel vs sequential equivalence)
- [ ] Security audit (external); continuous audit cadence post-mainnet
- [x] Documentation, runbook

4.4 Randomness & Fair Ordering

- [x] VDF/VRF for verifiable randomness (leader selection; leader_from_vrf, dummy_vrf_output)
- [x] Replace or augment round-robin with VDF-driven selection (leader_from_vrf stub)

4.5 Security Standards (see SECURITY-STANDARDS.md)

- [x] DDoS resistance: RPC rate-limiting (RateLimitConfig, governor crate)
- [x] Disk persistence (chain + state via --data-dir)
- [ ] Connection management at P2P layer
- [ ] Formal verification pipeline for critical contracts (staking, DeFi, automation)
- [ ] Bug bounty program
- [x] Incident response runbook (RUNBOOK.md §6)
- [ ] Post-quantum cryptography research and integration path
- [x] HD wallet path type (HdPath in boing-primitives)

4.6 Protocol Quality Assurance (QA) — see QUALITY-ASSURANCE-NETWORK.md

- [x] **boing-qa crate:** Rule types (Allow / Reject / Unsure), central rule registry, deterministic checks (bytecode size, opcode whitelist, well-formedness, purpose categories including meme/community/entertainment)
- [x] **Known edge-case resolution:** Implement §11 mappings so automation handles meme leniency, ambiguous declaration, new/unknown pattern; blocklist support
- [x] **Node integration:** Run QA at submit and/or mempool insert for ContractDeploy; structured rejection (rule_id, message); optional `boing_qaCheck` RPC
- [x] **Execution defense in depth:** QA check in `execute_contract_deploy` before `set_contract_code`
- [x] **Community QA pool:** Pending QA queue (`boing_qa::pool`), pool members, voting, max time T, default outcome at expiry; on-chain integration pending
- [ ] **Governance of rules:** Add/modify rules via time-locked governance; versioned rule sets

- [x] **Purpose in payload:** `ContractDeployWithPurpose` variant; `check_contract_deploy_full` with RuleRegistry
- [x] **Always-review categories:** Policy categories that force Unsure; configurable via RuleRegistry
- [x] **Scam patterns:** Byte-sequence blocklist (legitimacy heuristic) in RuleRegistry
- [x] **Soft rules:** Example: "other" + minimal description → Unsure
- [x] **Additional enhancements (optional):** Pre-flight SDK/CLI (boing_qaCheck RPC + SDK `client.qaCheck()`), public QA metrics, appeal path, canonical malice definition, deployer checklist (in progress)

Phase 5: UX & Human-Centered Innovations (Weeks 41–52)

5.1 Frictionless UX

- [x] Transaction simulation API (boing_simulateTransaction)
- [x] Gasless transactions (paymaster / meta-tx types: PaymasterConfig, SponsoredTransaction)
- [x] Human-readable signing (display_for_signing on SignedTransaction, SignedIntent)
- [ ] Transaction simulation API (pre-flight)
- [ ] Pre-confirmation / outcome guarantees where feasible
- [ ] Automatic chain routing (cross-chain UX)

5.1b Intent-Based Execution (see DEVELOPMENT-AND-ENHANCEMENTS.md)

- [x] Intent signing format (SignedIntent, IntentPool, boing_submitIntent RPC)
- [ ] Meta-router: orchestrate optimal cross-chain path
- [ ] Solver/executor integration for intent fulfillment

5.2 Trust & Verification

- [x] Intent-based transaction format (SignedIntent, boing_submitIntent)
- [ ] Simulation service / SDK
- [ ] Contract/address attestation hooks
- [ ] Verified display for addresses and amounts

5.3 Recoverability

- [x] Social recovery types (Guardian, RecoveryRequest in boing-primitives)
- [ ] Protocol-level recovery flows
- [ ] Optional MPC / threshold custody integration points

5.4 MEV & Cost Predictability

- [ ] VDF-based fair ordering (or equivalent)
- [ ] Encrypted mempool options
- [ ] Predictable gas model
- [ ] MEV-aware transaction design

5.5 Developer Experience

- [x] Boing CLI (`boing init` , `boing dev` , `boing deploy`)
- [x] RPC API specification (RPC-API-SPEC.md)

- [ ] Local/devnet parity for testing
- [ ] Cross-chain simulation support

5.6 Success-Based dApp Incentives & Boing SDK

- [x] dApp incentive formula (DappMetrics, dapp_incentive, VALUE_CAP_PER_DAPP)
- [ ] Per-dApp owner cap (governance parameter)
- [ ] Success metrics (tx count, fees, volume)
- [ ] Automated payout distribution
- [ ] Dynamic fee allocation & developer royalties (dApp-specified splits)
- [ ] Reputation-based resource access (priority processing, discounted gas)
- [ ] Boing SDK (`boing init` , `boing dev` , `boing deploy`)
- [x] Metrics registration API

5.6b Boing Studio & AI-Assisted SDK

- [ ] Boing Studio IDE (Remix-style: templates, debugging, one-click deploy)
- [ ] AI-assisted SDK (code generation, vulnerability scanning, optimization)

5.7 Decentralized Automation

- [x] Native scheduler (CronSchedule, Scheduler)
- [x] Execution verification (ExecutorAttestation, ZkpProof, FraudProof, OracleAttestation)
- [x] Trigger-based execution hooks
- [x] Decentralized executor incentives (executor_reward, executor_slash in boing-automation)
- [ ] User-facing automation SDK

Phase 6: Technical Innovations for Ecosystem (Weeks 53–64+)

6.1 Cross-Chain Primitives

- [ ] Light client verification at protocol level
- [ ] Bridge standards (IBC-style or custom)
- [ ] Cross-chain messaging primitives
- [ ] Trust-minimized bridge design (ZKP/MPC relayers with slashing)
- [ ] Decentralized oracle networks for external data

6.2 Stateless & Light Clients

- [x] Verkle proof generation and verification (MerkleProof)
- [ ] Light client sync protocol
- [ ] Mobile-friendly validation
- [ ] Compact proof specs (for ecosystem adoption)
- [ ] Browser-based Wasm light clients

6.3 Intent & Advanced Execution

- [x] Intent signing format (SignedIntent, IntentPool, boing_submitIntent)
- [ ] Solver/executor integration points
- [ ] Pre-confirmation API

- [ ] Commit-reveal for MEV-sensitive ops

6.4 Compliance & Real-World

- [ ] Privacy-preserving attestation design
- [ ] Off-ramp / fiat integration points
- [ ] Payment rail primitives
- [ ] Compliance-friendly APIs (selective disclosure)

6.5 Open Standards

- [x] RPC/API spec (RPC-API-SPEC.md)
- [ ] Bridge protocol documentation
- [ ] Reference implementations for interoperability

6.6 Decentralized Storage & Network Monitoring

- [ ] Filecoin/Arweave integration for permanent archival (chain state, tx logs)
- [ ] SDK tools for IPFS/Filecoin dApp frontend deployment
- [ ] Network topology monitoring (node distribution, decentralization dashboards)

Phase 7: Testnet & Mainnet (Ongoing from Phase 4)

7.1 Public Testnet

- [ ] Public testnet launch
- [ ] Testnet incentive program (sustainability-first): use INCENTIVIZED-TESTNET.md for readiness checklist, incentive design (validators, developers, users), and 2–4 week duration.
- [ ] Validator onboarding
- [ ] boing.finance integration (testnet)
- [ ] Community engagement: grant programs, hackathons, educational content

7.2 Mainnet

- [ ] Token distribution (fair launch)
- [ ] Bridge to boing.finance and other chains
- [ ] Mainnet launch
- [ ] Governance activation

Innovation Summary (from Design Plan)

Category	Key Items
UX	Native AA, gasless, human-readable signing, simulation, social recovery
Trust	Intent display, attestation, verified addresses
MEV	VDF ordering, encrypted mempool, predictable gas
Technical	Verkle stateless clients, light-client bridges, intent architecture
Ecosystem	Open standards, bridge protocols, reusable specs

Sustainability	Emission curve, fee-dominant revenue, no reward cliffs

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Task Checklist (Copy for Tracking)

```
Phase 1: Foundation
[x] Ed25519 tx signing/verification
[x] HotStuff propose → vote → commit
[x] Consensus unit tests
[x] Primitives + scheduler tests
[ ] State root (Merkle of accounts)
[ ] VM: gas stub, nonce check
[ ] Scheduler: parallel batch execution

Phase 2: Networking
[ ] libp2p swarm + gossip
[ ] Block/transaction propagation
[ ] Sync protocol
[ ] Node: mempool, block production
[ ] Multi-node testnet

Phase 3: Verkle & VM
[ ] Verkle tree implementation
[ ] Custom VM bytecode + interpreter
[ ] Full parallel execution

Phase 4: PoS & Hardening
[ ] PoS validator set
[ ] Native AA, adaptive gas, phased governance
[ ] Security audit

Phase 5: UX & Human-Centered
[ ] Gasless, simulation, human-readable signing
[ ] Social recovery
[ ] MEV mitigation
[ ] Unified SDK

Phase 6: Technical Innovations
[ ] Light client bridges
[ ] Intent architecture
[ ] Open standards
[ ] Compliance rails
```

Boing Network — Authentic. Decentralized. Optimal. Sustainable.